ZULFA FATAH AKBAR AHMAD

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LinkedIn | GitHub | Semarang, Indonesia (Open to relocation / remote)

PROFILE

Software Engineer with 3+ years of experience building web applications with React, Next.js, and TypeScript, and recent hands-on experience in C# and ASP.NET Core on Rock Imager, software that controls protein crystallization screening instruments at Formulatrix. Comfortable across the full stack, from real-time UIs to backend services, unit testing, and end-to-end test automation.

KEY SKILLS

Languages: C#, TypeScript, JavaScript, SQL

Backend & Databases: ASP.NET Core, Entity Framework Core, SignalR, REST APIs, MySQL, SQLite, Azure Cosmos DB

Frontend: React, Next.js, React Native, Redux, Zustand, Context API, TailwindCSS

Testing: NUnit, Playwright, mutation testing

Practices: SOLID, Clean Architecture, dependency injection, hardware/robot control, task scheduling, code review

Tools: Git, GitHub, GitLab, Bitbucket, Jira, Figma, OpenProject

EXPERIENCE

Software Engineer | Formulatrix August 2026 – November 2026

Rock Imager: software that controls protein crystallization screening instruments

- Developed the "Duplicate Plate Configuration" feature (React, TypeScript, C#, ASP.NET Core), eliminating the manual export/import step users needed to copy plate configurations.
- Resolved 10+ bugs in Rock Imager, including fixing a connection-loss defect that let users operate the instrument while its Rock Maker integration was unreachable, by adding a blocking connection-lost modal.
- Wrote 50+ NUnit tests for the Plate Storage Retriever (PSR, the robotic arm that moves plates between the load port, storage, and imager) and the Task Scheduler (robot task queueing, execution, and state detection), raising coverage from 1% to 80% for the two modules.
- Applied mutation testing with Stryker.NET to validate test quality on critical logic.
- Built Playwright end-to-end tests for 5 critical Rock Imager workflows, covering plate transfers between the load port, storage, and imager, optics and camera control, and manual imaging, automating regression checks that were previously done by hand.

Front-end Developer | PT. Tiga Daya Digital Indonesia January 2025 – December 2025

- Developed and maintained core features of an online auction platform for JBA Indonesia (lelang.jba.co.id) using React, Next.js, TypeScript, Socket.io, TailwindCSS, and Zustand, supporting real-time bidding for 50+ concurrent bidders at peak.
- Built VISION, an admin dashboard with modular-monolith architecture, complex data tables, real-time data handling, and full CRUD; created and documented 10+ modular components in Storybook.
- Integrated 20+ REST API endpoints with efficient data fetching, caching, and error handling.

Front-end Developer | PT. Bukit Vista Nusantara July 2022 – August 2024

- Designed and developed web application interfaces using React, Redux, Context API, Material UI, and TailwindCSS for an internal platform managing 100+ properties.
- Fixed critical bugs, optimized rendering logic and initial load.
- Conducted code reviews and mentored 2 interns to maintain code quality and best practices.
- Deployed features to production, collaborating closely with backend, design, and product teams.
- Contributed to a WordPress (PHP) website and a small Flutter mobile application.

PROJECTS & TRAINING

C# & ASP.NET Core Bootcamp | Formulatrix June 2026 – July 2026

- Developed LunchSpot, a real-time canteen table monitoring system using a YOLO object detector, ASP.NET Core, SignalR, and React.

- Built a Motorcycle Parking Management API using ASP.NET Core, Entity Framework Core, ASP.NET Core Identity, and SQLite.
- Applied SOLID principles, Clean Architecture, dependency injection, and asynchronous programming across backend projects.

EDUCATION

Diponegoro University August 2019 – August 2023

Bachelor of Engineering (S.T.) in Computer Engineering | GPA: 3.88/4.00

CERTIFICATION

- CCNA: Introduction to Networks, Cisco